

Mitigating Anchoring Bias in Long-Term Auditor Engagements

Complete Study Overview — Highlights, Shortcomings & the Path Forward

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Companion overview to the full manuscript (Research_Paper_YMalik_FULL_DRAFT.pdf).
Shortcomings marked **in red**.

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1. The Study at a Glance

Element	Detail
Research question	Which organizational interventions reduce anchoring bias in long-term (multi-year) audit engagements, and through what mechanisms?
Theoretical lens	Dual-process theory; anchoring-and-adjustment heuristic (Tversky & Kahneman, 1974)
Design	Cross-sectional survey; 55-item Likert instrument; IRB-25-0462
Model	8 interventions → 2 mediators → 1 outcome (11 constructs, 16 hypotheses)
Scoped analysis	EFA (PAF, Direct Oblimin) + reliability; CFA/regression/PLS-SEM specified for the next stage
Achieved outcome	Pilot / feasibility study — instrument fielded and proven; required analyses could not be performed at the achieved sample

Element	Detail
Manuscript	~16,200 words, Ch. 1–6 + references + Appendix A — complete

2. The Research Model — All 11 Constructs

The eight interventions (independent variables):

#	Construct	What it measures	Sample item
1	Training & Awareness (TA)	Firm training on cognitive biases and concrete de-anchoring techniques	"My firm provides specific training on cognitive biases (such as anchoring) that can affect audit judgments."
2	Rotation of Auditors (RA)	Personnel rotation bringing a fresh look to recurring engagements	"Rotation in my firm is frequent enough to bring a 'fresh look' to recurring clients."
3	Use of Analytical Tools (AT)	Data analytics generating independent, current-period expectations	"I routinely use data-analytics tools that flag anomalies in client data."
4	Structured Auditing Processes (SAP)	Standardized procedures forcing evidence-before-anchor evaluation	"Procedures require me to evaluate current evidence before consulting the prior-year conclusion."
5	Feedback & Reflection (FR)	Debriefs and judgment-quality feedback loops	"Post-engagement debriefs prompt me to reflect on judgment errors."
6	Independent Reviews (IR)	Substantive re-examination by uninvolved reviewers	"Reviewers challenge whether my judgments rely too heavily on prior-year audit conclusions."
7	Regulatory & Professional Guidance (RPG)	PCAOB/AICPA/IIA/COSO standards shaping judgment discipline	"The possibility of inspection encourages me to double-check the support behind my judgments."
8	Performance Metrics & Incentives (PMI)	Evaluation systems rewarding judgment quality over speed	"My performance evaluation rewards judgment quality, not only efficiency."

The two mediators and the outcome:

#	Construct	Role	What it measures
9	Auditor Judgment Quality (AJQ)	Cognitive mediator	Careful, objective evaluation of independent evidence
10	Audit Process Rigor (APR)	Procedural mediator	Thorough, consistent, disciplined execution of the audit process

#	Construct	Role	What it measures
11	Reduction in Anchoring Bias (RAB)	Outcome	Final judgments driven by current evidence, not initial reference points

Each construct: **5 Likert items** (1 reverse-coded) → **55 substantive items** total, plus 5 eligibility screens and 2 attention checks.

3. The 16 Hypotheses in Brief

- **8 direct paths (H1, H3, H5, H7, H9, H11, H13, H15):** each intervention → reduced anchoring bias.
- **8 mediated paths (H2, H4, H6, H8, H10, H12, H14, H16):** the six judgment-oriented interventions operate through **AJQ** (cognitive pathway); the two structural interventions (RA, SAP) operate through **APR** (procedural pathway).
- Status: fully developed in Ch. 3 (one section per hypothesis); **reserved for future structural testing — not testable at the achieved sample.**

4. The Instrument & Quality Architecture

Built in Qualtrics (v2.1 post-pilot, fielded July 1, 2026): 5 eligibility screens (US · English · audit role ≤24mo · continuing-engagement experience · attestation) → 55 items in 11 blocks → 2 embedded attention checks → open-ended anchoring-experience item → demographics. Anonymized, bot-detected, duplicate-protected. **Proven in the field: zero attention-check failures among complete respondents; platform response-quality index 95%.**

5. The Study Journey — Timeline of What Actually Happened

When (2026)	Event
May 20	Advisor strategy meeting: keep IRB-approved model, EFA scope, "close that book"; AI saved for dissertation
May 30	Ch. 2/3 structure set with advisor (11 construct sections, 16 hypothesis sections)
Jun 30	Advisor directive confirmed: EFA + reliabilities only; PAF; Direct Oblimin; target n=100
Late Jun	Pilot via LinkedIn contacts → wording refinements → v2.1
Jul 1	Survey live in Qualtrics; organic outreach (LinkedIn, WhatsApp, direct)
Jul 10–15	Paid panel funded (~\$1,000, 100 places, \$28/hr effective) and published
Jul 1–22	Collection trickle: the eligible population barely exists on research panels
Jul 22	Data exported and screened by scripted protocol → only 4–5 usable responses
Late Jul	Resolution meeting: pass/fail path accepted by university; Ch. 1, 4–6 rewritten; simulated-data exercise to follow separately

6. Highlights — What This Study Achieved

- 1 **A complete, theory-grounded 11-construct model** with 16 hypotheses — intact and reusable.
- 2 **A fielded, field-proven instrument** — clean administration, working screens/checks, high response quality.
- 3 **A reproducible data pipeline** — scripted 9-step cleaning protocol, exclusion logging, version control.
- 4 **A contract-grade methodology** (Ch. 4) — EFA→reliability→CFA→regression→PLS-SEM with explicit decision criteria at every step.
- 5 **The decisive feasibility finding, platform-verified:** the panel's own screening found **only ~20 eligible participants out of 334,976** (~6 per 100,000) — proof the constraint was population prevalence, not design, effort, or price.
- 6 **A scholarship-grounded dissertation bridge** — the AI-in-audit extension anchored in 10 verified top-tier sources (JAR, RAST ×3, Accounting Horizons, JIS, IJAIS, Scientific African, PCAOB, IAASB).
- 7 **Full transparency** — every decision, directive, and result documented in a public, timestamped repository.

7. Shortcomings — Stated Plainly

- **The required analyses were not performed. EFA, reliability, CFA, regression, and PLS-SEM were all infeasible at the achieved sample — the study validates nothing statistically.**
- **The achieved sample was 4–5 usable responses against a target of 100, from 23 raw starts — a >95% shortfall.**
- **The population definition was fatally narrow: U.S. auditors with current/recent roles and multi-year engagement experience ≈ 6 per 100,000 panel participants. The study was unfillable as designed on open panels.**
- **~\$1,000 of paid recruitment yielded almost nothing — an effective cost per usable panel response orders beyond any planned rate.**
- **Data collection started too late in the term — the recruitment risk surfaced with no runway left to re-scope within the course window.**
- **Self-report design limitations (social desirability, limited self-insight on one's own bias) remain unaddressed pending the method-diversified follow-up.**

8. The Pivot: From "U.S. Auditors" to "Risk & Assurance Professionals"

The arithmetic that forces the pivot: $20 \text{ eligible} \div 334,976 \text{ panelists} \approx 0.006\%$. No budget fills 100 seats from a pool of 20. To collect a factor-analysis-grade sample, the population definition must widen — without breaking the construct logic.

Why "risk & assurance" is the right widening (not a compromise): every construct in the model describes judgment discipline under recurring evidence — none is audit-license-specific. Anchoring on prior-period figures operates identically for:

Expanded population segment	Same anchoring mechanism because...
Internal auditors & SOX/controls testers	Prior-year workpapers and known-issue lists anchor current testing
Enterprise risk management professionals	Prior risk ratings anchor current risk assessments
Compliance & regulatory-review staff	Prior findings anchor current review scope
Credit review / loan review officers	Prior credit grades anchor current grading
Quality assurance & controls monitoring	Prior defect baselines anchor current sampling
Financial analysts performing recurring reviews	Prior-period actuals/budgets anchor current expectations

What changes: screener wording ("audit-related role" → "risk, audit, compliance, or controls role with recurring multi-period reviews"), the population section of the methods, and — **required first — an IRB amendment (the approved protocol specifies U.S. auditors)**. **What does not change:** the 11 constructs, the 55 items (minor wording generalization at most), the 16 hypotheses, the analysis contract. Estimated prevalence effect: risk/compliance/finance professionals are plausibly **50–100× more prevalent** on research panels than practicing auditors — moving the eligible pool from ~20 into the thousands, at which point the recovered ~\$980 funds a real sample.

9. What Carries Forward to the Dissertation

The entire apparatus — model, instrument, pipeline, analysis contract — plus the AI extension: **do AI/LLM audit tools attenuate human anchors, or substitute algorithmic anchors of their own?** (Algorithm aversion: Commerford et al., 2022. Algorithmic anchors: Koreff, 2022; Kokina et al., 2025. Capability: Eulerich et al., 2024; Emmett et al., 2025. Outcomes: Fedyk et al., 2022. Governance: PCAOB, 2024; IAASB, 2024.) The broadened risk-professional population is where AI decision aids are deploying fastest — the pivot and the AI extension reinforce each other.

10. Immediate Next Steps

- 1 **Submit the manuscript** (Friday target; Tuesday hard deadline).
- 2 **Complete the advisor's simulated-data exercise** (separate from the paper) — Jamovi runbook ready.
- 3 **Stop the paid panel study & recover ~\$980** (submissions already approved/paid).
- 4 **File the IRB amendment** for the broadened risk & assurance population.
- 5 **Relaunch collection under the dissertation** with corrected screeners — see `Prolific_Relaunch_Guide.md`.